Scaling Public-Private Partnerships in Ireland

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Glossary

Acronyms

AIRAP	Accelerating Infrastructure Report and Action Plan
ESA 2010	European System of Accounts 2010
FDI	Foreign Direct Investment
GDP	Gross Domestic Product
IMF	International Monetary Fund
IPSAS	International Public Sector Accounting Standards
IRR	Internal Rate of Return
MDD	Modified Domestic Demand
NER	Nominal Exchange Rate
PFRAM 2.0	Public-Private Partnership Fiscal Risk Assessment Model
PPP	Public-Private Partnership
PSC	Public Spending Code
VfM	Value for Money

Key Terms

Availability payment	A regular government payment to a private partner for keeping an asset available at agreed standards.
Contingent liability	A possible future payment that only arises if a specific event occurs.
Correlated risk	Risk that moves together across several projects.
Diversification	Reducing overall risk by combining projects exposed to different pressures.
Fiscal exposure	The financial obligation or risk borne by the State.
P50 / P95 / P99	Percentile outcomes: median, severe, and very severe scenarios.
Portfolio approach	Assessing projects together rather than one by one.
Stress test	A modelled adverse scenario used to test resilience.
Tail risk	The risk of rare but severe losses.
Value for Money	An assessment of whether a proposed public investment is justified by its expected costs, benefits and risks.

Executive Summary

Ireland has committed to a step-change in public infrastructure delivery to support population growth, climate targets and long-term competitiveness. Delivering this programme will require not only higher capital spending, but a delivery model capable of accelerating projects while maintaining fiscal control.

Public-Private Partnerships (PPPs) are an established framework capable of mobilising private finance efficiently. This report argues that scaling PPP deployment will be one important vehicle for achieving the envisioned level of capital spending and ensuring private sector involvement translates efficiently into quality public infrastructure.

Expanding PPP use increases the complexity of assessing the State's fiscal exposure. The report therefore argues for a centralised portfolio approach to PPP selection, monitoring and fiscal risk assessment.

The report develops a blueprint for portfolio-level management, illustrated by a proof-of-concept case study. The approach builds on a model developed by the World Bank and IMF, to incorporate risks shared across projects. A stress-testing approach is adopted to properly quantify the portfolio's fiscal exposure in adverse macroeconomic conditions. The case study demonstrates portfolio monitoring in action, showing how modelling can identify diversifying projects and support strategic sequencing.

The research and case study inform the policy recommendations in box 1 and are expanded on further in the policy actions section of the report. The recommendations are aligned with the government's current strategy, as outlined in the Accelerating Infrastructure Report and Action Plan.

PPPs can help Ireland deliver infrastructure at scale, but only if their risks are managed collectively. The State should therefore move beyond isolated project appraisal and adopt a portfolio approach that tests how each new PPP affects aggregate fiscal exposure, downside risk and strategic direction.

Box 1: Policy Actions / Recommendations

- **Establish a central PPP risk oversight function.** The function will be embedded within the existing central PPP oversight unit, expanding its mandate to include portfolio risk ownership. The function will be responsible for maintaining a live model of the PPP portfolio, utilising and enhancing the proof-of-concept developed in this report to accurately quantify fiscal exposure.
- **Embed portfolio stress-testing within the Public Spending Code**. The assessment and procurement framework must be updated to account for aggregate fiscal exposure from PPP projects. Portfolio level analysis can be inserted prior to project tendering, without a major redesign of the code.
- **Select and monitor projects using a portfolio approach**. Expand the mandate of the central oversight unit to include project selection. This will ensure projects are strategically aligned, and fiscal exposure is adequately diversified. A comply-or-explain framework is recommended, leaving final decision-making with government ministers to ensure departmental autonomy.

Contents

Introduction

Key Messages

- Significant infrastructure investment is needed in Ireland to meet its climate objectives, maintain competitiveness, and accommodate shifting demographics.
- Ireland has a weak track record of delivering large infrastructure projects on-time and within budget.
- The Accelerating Infrastructure Report and Action Plan (AIRAP) outlines an ambitious programme of infrastructure investment that necessitates higher levels of private sector involvement, and tighter oversight.

The Infrastructure Imperative

Ireland is committed to accelerating public infrastructure delivery to sustain its growing economy, meet its climate objectives, and continue to attract investment. The government has announced an unprecedented capital investment package of €275 billion by 2035 (Department of the Taoiseach, 2025).

This commitment is timely, as demographic changes will build pressure on existing infrastructure over the next decade. Recent projections from the Economic and Social Research Institute forecast population growth of 17% between 2022 and 2040, with a growing share aged 65 and older (Bergin et al., 2026). Accommodating these changes will require investment across housing, health, transport, and utilities infrastructure.

Coinciding with this pressure is an opportunity to improve competitiveness and maintain the country's impressive track record of Foreign Direct Investment (FDI). EY's Attractiveness Survey (2024) identified Transport and Energy investment as two of the top five policy focus areas to increase competitiveness. The same report noted that Ireland has slipped in the competitiveness league table, underlining the urgency of action.

Ireland's patchy track record of infrastructure delivery is partly to blame for the current scale of investment needed. Delivery timelines are worsening, with a recent government report noting that many project development cycles have doubled in length over the last 20 years (Department of Public Expenditure, 2025). A pointed example is the North-South Interconnector (NSIC), an electricity line that will facilitate the connection of 900

MW of renewable energy. The project was launched in 2007 and is now expected to be complete by 2031, a staggering duration of 24 years (Department of Climate, 2024). Many projects have also been troubled by significant cost escalation, including a greater than two-fold cost increase in delivering the National Children's hospital (Department of the Taoiseach, 2019).

Government Strategy

The Accelerating Infrastructure Report and Action Plan (2025) codifies the government's intention to meet this challenge by reforming and scaling infrastructure delivery. Committing to €110 billion of new public capital investment by 2030, the report details 30 actions designed to restructure delivery at legal, regulatory, and implementation levels.

> ***"Government must be prepared to step up to the plate and to state that sometimes, we are prepared to accept a higher degree of project risk in exchange for a faster delivery cycle."***
>
> – Accelerating Infrastructure Report and Action Plan, Ministerial Foreword

Pillar 3 of the AIRAP groups actions designed to address coordination and delivery of projects. A welcome component of this pillar is the more active role the Department of Public Expenditure, Infrastructure, Public Service Reform and Digitalisation is tasked with taking. The department will maintain central oversight of projects, ensuring alignment between planning, funding, and delivery, as well as tracking performance, risks, and interdependencies. This has the potential to deliver stronger fiscal control of projects and unlock strategic synergies across the portfolio (Coyle, 2022).

Actions 15 and 25 of the AIRAP prioritise private sector involvement and risk sharing, recognising the efficiency and innovation benefits private finance can deliver.

The strategy is directionally strong but leaves practical questions unresolved. While stronger coordination and risk oversight is called for, a framework to implement this is unspecified. In addition, although private sector participation and greater risk sharing are recognised, little attention is given to the fiscal controls required to scale private sector involvement. This gap between strategic ambition and operational implementation provides the starting point for this report.

The Argument

This paper argues that enhanced strategic coordination is necessary to scale infrastructure delivery and achieve Ireland's national ambitions.

New private sector partnerships will be required to unlock capital and deliver infrastructure efficiently. It is argued that centralisation of project selection and risk management are necessary to manage the tension between public and private sector interests.

This report outlines a pathway towards effective private sector mobilisation, focusing on the AIRAP priorities of co-ordinated delivery and risk-sharing, two areas which have historically proven difficult.

Section 1 reviews private sector participation in Ireland's public infrastructure, arguing that reliance must shift away from ad hoc consulting, towards comprehensive Public-Private Partnership (PPP) contracts. Section 2 introduces the portfolio approach to PPP management. Shared risks between infrastructure projects pose a material fiscal threat, which can be managed effectively through central oversight. Sections 3 and 4 illustrate the approach using a stylised synthetic analogue of Ireland's transport PPP portfolio. The fiscal impact of shared transport risks are modelled by adapting the World Bank's Public-Private Partnership Fiscal Risk Assessment Model (PFRAM 2.0). Section 5 translates findings into phased policy reforms.

1. Mobilising the Private Sector

Key Messages

- Project delivery is heavily reliant on private sector consultants, inflating cost and missing an opportunity to create consistency across projects.
- Internalising expertise within the public sector may reduce costs but lower private sector involvement could undermine efficiency.
- PPPs offer a more structured approach to risk allocation and fiscal oversight than ad hoc alternatives, though their value for money depends critically on robust contract design and portfolio management.

Moving Beyond Consultancy Dependence

Private sector involvement in public infrastructure delivery takes a variety of forms across the project cycle. At a minimum in Ireland, private sector consultants are relied on heavily to provide technical support and advisory services to projects. The government has recently come under fire for this dependence, and a push to bring expertise in-house is reflected in the AIRAP (O'Donnell, 2025). The ad hoc nature of consulting brings elevated costs, as well as a lack of consistency and coordination between projects. The concern is not simply budgetary. Mazzucato and Collington (2023) argue that persistent reliance on consultants can deplete the State’s own knowledge base, leaving public bodies less able to specify, challenge and manage complex projects over time.

Internalising expertise brings trade-offs, particularly while the state is being accused of ‘government-by-agency’, reflecting the increasing importance of public bodies (Ó Ceallaigh, 2025). The government is balancing this risk by creating a centralised coordination body, thereby increasing expertise while maintaining direct political oversight.

While the move towards centralised public expertise should be welcomed, reduced private sector involvement risks leaving benefits unrealised. The profit incentive of private companies drives efficiency and innovation when correctly utilised. Considering the scale of planned investment, capturing private sector efficiencies will be crucial to the programme’s success.

Public-Private Partnerships

One approach to private sector involvement that avoids the pitfalls of ad hoc consultancy is the use of Public-Private Partnerships (PPPs). These are long-term contracts between private companies and public bodies to finance, build, and operate public infrastructure. Widening the scope of private involvement under a single contract optimises lifecycle risk sharing, while maintaining oversight of delivery and operation (OECD, 2008).

A further benefit is the reduced upfront cost of capital projects through private sector financing. Although PPPs reduce the upfront project cost, the overall cost to the taxpayer is similar compared to traditional procurement, reflecting the higher cost of private finance relative to government borrowing (International Monetary Fund, 2016). This creates a risk of fiscal illusion, as long-term government exposure is obscured (Irwin, 2012). The evidence on PPP value for money is also mixed. The United Kingdom's experience with Private Finance Initiative contracts found that inflexibility and financing costs frequently offset gains (National Audit Office, 2018). These risks underline the importance of robust fiscal oversight.

The scale and length of PPP contracts entail significant risk. Each project risk is allocated to the partner who is best placed to handle it. For example, the public partner typically retains regulatory risk, while the private partner typically shoulders design and construction risk. Risks can also be shared between the public and private partners. For example, demand risk is often shouldered by the private partner, with contingent liabilities for the public partner if demand falls below an agreed threshold. PPP contract design must also comply with EU state aid rules. Demand support mechanisms and other guarantees may confer advantages not available on market terms, requiring careful structuring within the European Commission's General Block Exemption Regulation framework on a project-by-project basis.

Concerningly in Ireland, technical guidance is largely silent on the combined risk of PPP projects when viewed at the portfolio level. Many fiscal risks and contingent liabilities are exposed to the same underlying economic pressures, meaning that problems affecting one project may also affect others at the same time. However, there is little consideration of whether a new PPP project would increase the government's exposure to these shared risks or help diversify overall risk across the portfolio. This gap is explored in the context of Ireland's PPP framework in the following section. Before doing so, it is worth considering PPPs within the broader spectrum of procurement options available to government.

Competing Procurement Options

Public-Private Partnerships are one option to move beyond consultancy dependence. A wider spectrum of delivery models exists, each with merit in the right context. Box 2 explores alternative strategies to scaling Public-Private Partnerships.

Box 2: Procurement Reform Options

- **Maintaining the Status Quo** is not a viable option. The AIRAP acknowledges that reform is needed to improve Ireland's poor track-record of infrastructure delivery.

- **Reforming traditional procurement** is a key component of the AIRAP and must be acknowledged. Additionally, traditional procurement is often preferred for flagship projects, where decision-makers do not want to appear to be privatising public infrastructure. However, the scale of investment envisioned by the government requires additional private capital mobilisation.

- **Hybrid models** fill the gap between traditional procurement and PPPs, encompassing construction or management of the infrastructure project. These may be more suitable for smaller project that cannot justify the expense and risk of PPP contracts. However, they sacrifice the efficiency gained through a contract integrated across the full lifecycle of the project.

- **Enhanced project-level contracting** has the potential to improve project outcomes and is a viable area of focus. The problem here is that even a well-designed individual contract does not address the aggregate risks that arise when multiple projects share risk factors.

Traditional procurement and hybrid models retain a place in Ireland's strategy, and improved contract structuring is valuable across all models. However, PPPs address the gap of mobilising private capital and managing lifecycle delivery at scale. The government's infrastructure ambition implies this gap is becoming more important, therefore PPPs have a key place in the strategy.

Ireland's PPP Framework

Ireland currently has a portfolio of 31 active PPPs across Transport, Housing, Healthcare, and Education sectors (Public Private Partnership Unit, 2026). A Central PPP Unit develops the general policy framework, provides guidance to departments, and chairs a high-level steering committee overseeing PPP progress. Project selection and management is devolved to the relevant department or public body entering the contract. The National Development Finance Agency (NDFA) also plays an important supporting role, providing financial advice, procurement support, project delivery, and contract management services.

Determining whether a project is suitable for procurement via PPP entails a Value for Money (VfM) assessment, and benchmarking against conventional public procurement methods (Department of Public Expenditure, 2019). The VfM assessment evaluates project justification based on expected costs, benefits and risks, and the Public Sector Benchmark then compares PPP procurement against traditional public delivery. Projects must be sufficiently large to justify the high transaction costs associated with negotiating a PPP contract. Infrastructure with significant operational costs and relatively stable long-term demand is often better suited to the broader lifecycle scope of a PPP arrangement. In addition, there must be sufficient private sector interest to support a competitive procurement process.

The gap addressed in this report is therefore not institutional, but functional. Sheppard and Beck (2018) note that Ireland's PPP framework followed the UK model closely, while Connolly, Reeves and Wall (2009) argue that PPP adoption in Ireland can partly be explained by the tendency to copy accepted institutional models because they signal modernisation and legitimacy, termed mimetic isomorphism (DiMaggio & Powell, 1983). While Ireland's PPP institutions are not merely symbolic, they must be strengthened beyond guidance and procurement support, towards active portfolio-level selection and risk management. This matters because scaling PPP use safely depends on the State's ability to identify, price and allocate project risks consistently across the portfolio. The following section explores a new framework to select and assess projects at the portfolio level, in line with the AIRAP's focus on centralised delivery.

2. The Portfolio Approach

Key Messages

- Ireland's existing approach of project-level evaluation underestimates aggregate fiscal exposure and misses an opportunity to select projects strategically.
- Portfolio-level assessment addresses these issues and aligns with the AIRAP's goal of building centralised public sector expertise.
- An existing model developed by the World Bank and IMF provides a useful starting point for portfolio-level PPP management.
- Understanding shared risks across the PPP portfolio is crucial to assess aggregate outcomes. The basic model can be extended to account for this through correlated risk structures and simulated stress-testing.

From Project Selection to Portfolio Strategy

Ireland's current approach to infrastructure evaluation treats projects as largely independent investments. Procurement is managed individually through the Public Spending Code and Capital Appraisal Guidelines, overlooking how risks and benefits interact across the broader infrastructure programme (Department of Public Expenditure, 2019). The downside of this approach is two-fold:

- Policy makers must be concerned with the combined public exposure created by multiple PPP contracts. Without joint assessment of shared risks, the aggregate fiscal risk may be underestimated. The financial crisis of 2008 is an important example illustrating why this approach should be applied to PPPs. Twenty PPP projects were either postponed or cancelled as economic activity slowed, highlighting that multiple projects were exposed to the same macroeconomic risk (MacCarthaigh & McNamara, 2015). The portfolio stress-testing approach may have highlighted the joint-risk of PPP projects prior to the financial crisis, allowing policy-makers time to adjust project contracts to restructure aggregate risk.
- Assessing projects individually misses an opportunity to identify valuable synergies between projects. Infrastructure projects can often be enhanced through integration. For example, transport corridors can be developed in conjunction with electricity infrastructure to construct efficient electric vehicle

charging networks. Viewing infrastructure at the portfolio level is an opportunity to select projects more strategically, ensuring alignment with the wider portfolio.

The portfolio approach to state-level risk management is not new. A close analogue can be found in banking regulation, where the failure of supervision before the 2008 financial crisis demonstrated the limits of assessing institutions in isolation. In Ireland, the financial crisis exposed the need for a macroprudential approach. The Central Bank of Ireland now applies stress-testing frameworks to assess how macroeconomic shocks affect aggregate banking resilience (Morell et al., 2022). This provides a useful precedent for PPP governance. Where projects share exposure to common shocks, fiscal risk should be assessed at portfolio level rather than project by project.

International experience also validates the case for centralised portfolio risk management. Colombia's Ministry of Finance created a dedicated PPP risk unit following demand guarantee calls that cost the government approximately 2% of GDP, subsequently developing stochastic methodologies to model contingent liabilities across its PPP portfolio (The World Bank, 2020). Chile has adopted a similar approach through a dedicated Division of Contingent Liabilities and Concessions (Reyes-Tagle & Others, 2018).

The portfolio approach aligns tightly with the objectives of the AIRAP. The modelling approach can be implemented within the proposed central oversight unit of the Department of Public Expenditure, Infrastructure, Public Service Reform and Digitalisation.

Strategic Benefits of PPP Portfolio Management

Fiscal Oversight is the key benefit of the approach. Understanding portfolio level contingent liabilities brings a clearer understanding of aggregate fiscal risk, providing government with an opportunity to manage it effectively. The scale of investment planned within the AIRAP brings significant risk. Long-term management is necessary to ensure control of the irreversible investments planned under AIRAP.

Diversification effects can reduce the volatility of liabilities within the portfolio. By selecting projects exposed to different macroeconomic and operational risks, Ireland can reduce the extent to which fiscal pressures materialise simultaneously across projects. For example, Toll-road PPPs may complement Social Housing PPPs during a recession. Where Toll-road revenues may decline due to reduced demand, social housing demand is likely to remain stable.

Strategic Sequencing of projects is possible for a centrally coordinated portfolio. Staggering projects may be beneficial where shared risk is strong. This approach is useful where projects are competing for labour and materials, reducing inflationary pressure. Additionally, sequencing becomes important where strategic complementarities are identified. For example, transport infrastructure may facilitate housing development once complete. The existing framework fragments selection and implementation to the department or public authority level, reducing the ability to co-ordinate project timelines across the portfolio.

Strategic Vision is strengthened by the portfolio approach because projects can be selected according to their contribution to long-term national objectives. The portfolio model supports a more strategic approach to infrastructure delivery, moving beyond reactive investment towards an anticipatory approach. The approach enables backcasting, working back from desired future states rather than projecting forward (Dreborg, 1996).

Centralised Contract Expertise can be developed to support PPP procurement. This aligns with the AIRAP's goal of bringing expertise in-house. The clarity gained through the modelling framework below strengthens the public partner's position when entering PPP contracts.

Modelling PPP Portfolios

The Public-Private Partnership Fiscal Risk Assessment Model (PFRAM 2.0) developed by the World Bank and IMF is an existing tool used to assess macro-fiscal implications of PPP portfolios and inform decision-making.

The model determines the expected timing and magnitude of public payments, assesses the impact on debt, and estimates the potential impact of contingent liabilities. Results are presented in accessible graphs and tables to assist non-technical decision makers. Boxes 3 and 4 outline what the model can and cannot do, where relevant to this research.

Box 3: PFRAM 2.0 Capabilities

- Generate a fiscal risk matrix for a PPP project.
- Toggle projects within a portfolio to understand marginal impacts.
- Sensitivity Analysis of GDP, Inflation, and Nominal Exchange Rate shocks.
- Simulate contract termination.
- Ongoing portfolio management.

Box 4: PFRAM 2.0 Limitations

- The model supports a maximum of 30 PPP projects.
- Risks within and across projects are assumed to be uncorrelated.
- Procurement models cannot be assessed, i.e. PPP vs traditional procurement.
- The model cannot justify the economic or social relevance of a project.
- Sensitivity analysis uses GDP rather than Modified Domestic Demand (MDD), which is more appropriate for Ireland.
- PFRAM applies IPSAS-based economic-risk accounting principles, whereas Irish PPP's use the ESA 2010 statistical classification to determine whether projects remain on or off the general government balance sheet. The IPSAS approach is more useful for fiscal risk analysis because it captures the government's underlying economic exposure to PPP liabilities, regardless of their formal accounting treatment.

Limitations of the model are important to understand in the context of evaluation sequencing. Procurement methods and Value for Money must be assessed prior to conducting analysis using the PFRAM 2.0 model.

Most significantly, the model does not account for correlation among risks within a project or across the portfolio. To overcome this limitation, an extension of the model is proposed in the following section.

Modelling Shared Risk

The purpose of modelling shared risk is to identify where negative outcomes are likely to arrive together. When growth slows, inflation rises, or a contractor comes under stress, several projects can be affected at once. A variety of risk factors are shared or correlated across infrastructure projects. Table 1 illustrates some common risk factors across projects, and their effects on the portfolio.

Table 1: Risk Factors Shared Across Infrastructure Projects

Risk Factor	Projects Affected	Transmission Mechanism	Portfolio Effect
GDP Slowdown	Toll roads	Lower traffic demand	Contingent Liabilities Triggered
Construction Inflation	Roads, hospitals	Shared inputs	Contingent Liabilities Triggered
Contractor Distress	Projects with shared private partner	Counterparty failure	Simultaneous delay or termination
Interest-rate Shock	Hospitals, housing, availability roads	Availability Payments	Fiscal Burden
Energy Shock	Utilities, transport	Construction and operating costs	Contingent Liabilities Triggered

Construction materials common between projects are subject to the same inflation and supply-chain risks. Steel and copper inputs are prone to price volatility, affecting power and building construction projects. Similarly, energy inputs are sensitive to global supply shocks and are a common risk factor across all construction projects. Recent disruptions in the Strait of Hormuz have already been associated with widespread input cost increases across the construction industry (The Wall Street Journal, 2025).

Counterparty risk may also be shared across projects. Among Ireland's 31 PPP projects, several companies are a private partner on three or more projects. BAM, for example, appears across multiple projects in the portfolio. The construction firm is also the principal contractor on the National Children's Hospital which is undergoing significant delay and cost overrun. This demonstrates the importance of monitoring concentrated counterparty exposure at the portfolio level rather than assessing projects in isolation.

Finally, as demonstrated by the financial crisis of 2008, infrastructure projects are exposed to broader fiscal risks and demand shocks. These risks lend themselves directly to the similar portfolio monitoring techniques used by the central bank to monitor financial system health.

To model these shared risks in practice, practitioners identify individual project risks first and then define how they move together. A correlation structure is built across the main risk drivers, and possible outcomes are then simulated to assess their likelihood.

Building on this basic approach, copulas can be used to assess extreme downside states where risks move strongly together. Copulas allow practitioners to model tail dependence separately from marginal risks (Furman et al., 2016).

Stress-testing scenarios can be defined to simulate recessions and other economic shocks. Advanced frameworks use network-style contagion models to assess the possibility that one failure may trigger more.

In the following case study, a simple correlated risk overlay is developed to illustrate how the PFRAM 2.0 model can be extended.

3. Illustrative Case Study

This section develops a stylised case study to illustrate the utility of the PFRAM 2.0 framework for PPP portfolio management and the fiscal implications of correlated risk structures. Full deployment to Ireland's PPP portfolio would require more granular contractual data and model validation. The simulations should therefore be read as illustrative scenario analysis rather than predictive fiscal estimates.

A synthetic PPP portfolio is constructed and calibrated to the broad characteristics of Ireland's existing PPPs. Precise reconstruction is constrained by limited public disclosure of contractual information and is outside the scope of this report. Instead, a limited number of representative synthetic projects are developed to illustrate portfolio composition, fiscal exposure, and correlated risk within a controlled modelling environment.

The use of stylised synthetic portfolios is common in fiscal risk analysis and stress-testing exercises where the objective is to examine structural relationships and transmission mechanisms rather than produce project-level forecasts (Borio et al., 2012; Cebotari et al., 2009).

Stylised Portfolio Construction

Transport Infrastructure Ireland has published an extensive review of their portfolio of 15 PPP projects (Transport Infrastructure Ireland, 2018). The projects vary in scale and incorporate a variety of risk-transfer mechanisms, providing a useful basis for the synthetic portfolio. A representative portfolio of three synthetic projects is constructed, based on real analogues. Table 2 provides high-level details of the projects, with a comprehensive breakdown included in the appendix.

Contractual structure and cash flow modelling is simplified for the purposes of this report, where the central focus is risk transmission.

Synthetic Name	Real Analogue	Structure	Funding Source	Fiscal Channel	Baseline Role

Western Motorway	M6 Galway – Ballinasloe	Toll motorway	User-funded	Limited direct fiscal exposure	Pure demand risk concession
Southern Tunnel	Limerick Tunnel	Toll with guarantee	User-funded & contingent support	Demand guarantee	Contingent liability case
Eastern Ring Road	M50 Upgrade	Availability motorway	Government-funded	Annual service payment	Direct fiscal obligation

Table 2: Baseline Synthetic Portfolio

Analytical Framework

The modelling framework is described in figure 1. Three stages of analysis are conducted to illustrate the utility of the PFRAM 2.0 model and examine correlated risk structures[1]. A full breakdown of the modelling assumptions are included in the appendix.

[1] **The full PFRAM 2.0 model and Python modelling script used in this analysis are available for download here.**

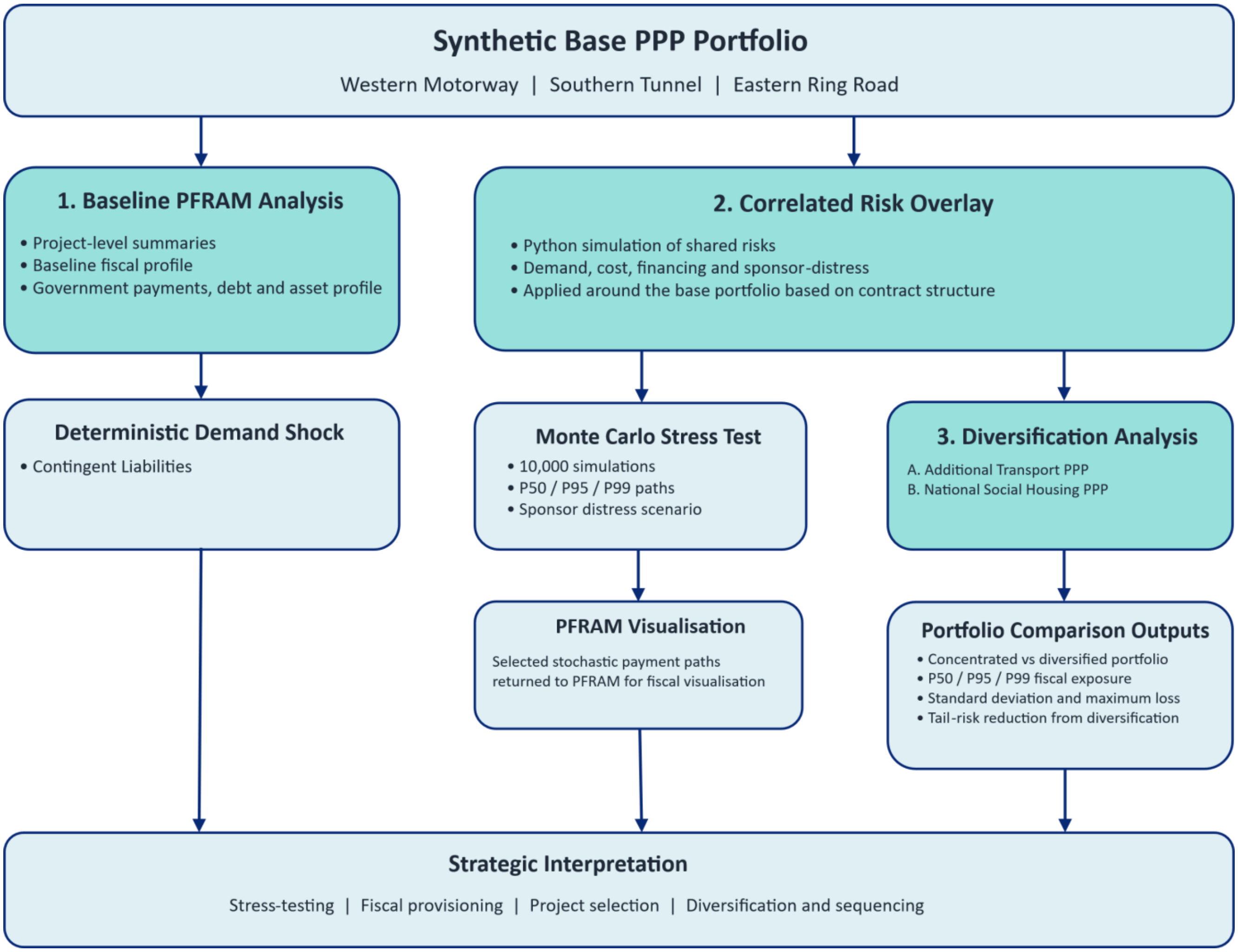

Figure 1: Three-Stage Modelling Framework

Correlated Risk Layer

A standalone Python script is developed to simulate the impact of stress-scenarios on government fiscal liabilities. The shared risks of the base portfolio are encoded in a stress-testing matrix, and Monte Carlo simulation is used to assess the fiscal impact distribution of a macroeconomic shock. A limited number of risks are considered to maintain illustrative clarity. The script outputs adjusted project cash flows, which can then be inputted to PFRAM 2.0 to visualise the shock using its existing functionalities.

Diversification Layer

The final stage of analysis illustrates portfolio diversification effects. Two possible projects are considered to extend the portfolio, as summarised in table 3. In the first case, a fourth transport project is added, concentrating the existing correlated risk structure. Alternatively, a Social Housing PPP is considered, weakening the correlation structure. The alternatives are compared on the basis of stress-testing simulations.

Table 3: Projects Considered to Extend the Base Portfolio

Synthetic Name	Portfolio Role	Structure	Funding Source	Baseline Role
Additional Transport PPP	Concentrated portfolio comparator	Availability motorway PPP	Government-funded	Highly correlated with the existing transport portfolio through demand, financing and sponsor-distress channels
National Social Housing PPP	Diversification project	Availability-based social infrastructure PPP	Government-funded	Exposed primarily to construction and lifecycle cost variation; limited exposure to transport-demand shocks

4. Results and Analysis

Baseline Portfolio Results

The baseline model establishes the deterministic fiscal profile of the synthetic PPP portfolio before risk shocks are introduced. Table 4 summarises key model outputs.

Table 4: Synthetic Base Portfolio Outputs. *The Equity Internal Rate of Return (IRR) indicates the project's viability from the private partner's perspective. The projects were constructed to generate moderate returns, highest for the Western Motorway where the private partner fully retains demand risk, and lowest for the Availability-based Eastern Motorway. Note, the baseline annual payment of €65 million for the Eastern Motorway refers to the first year of operation and is inflation linked.*

	Western Motorway	**Southern Tunnel**	**Eastern Motorway**	**Portfolio Total**
Private Partner Base Case Equity IRR	6.8%	6.0%	4.3%	N/A
Baseline Annual Government Payment *(€m, nominal)*	0	0	65	65
Total Government Cash Outflow *(€m, undiscounted)*	0	0	3,000	3,000

Direct government exposure in the base model is driven entirely by the availability payments of the Eastern Motorway project. Figures 2 and 3 illustrate project and government cash flows for the Eastern Motorway.

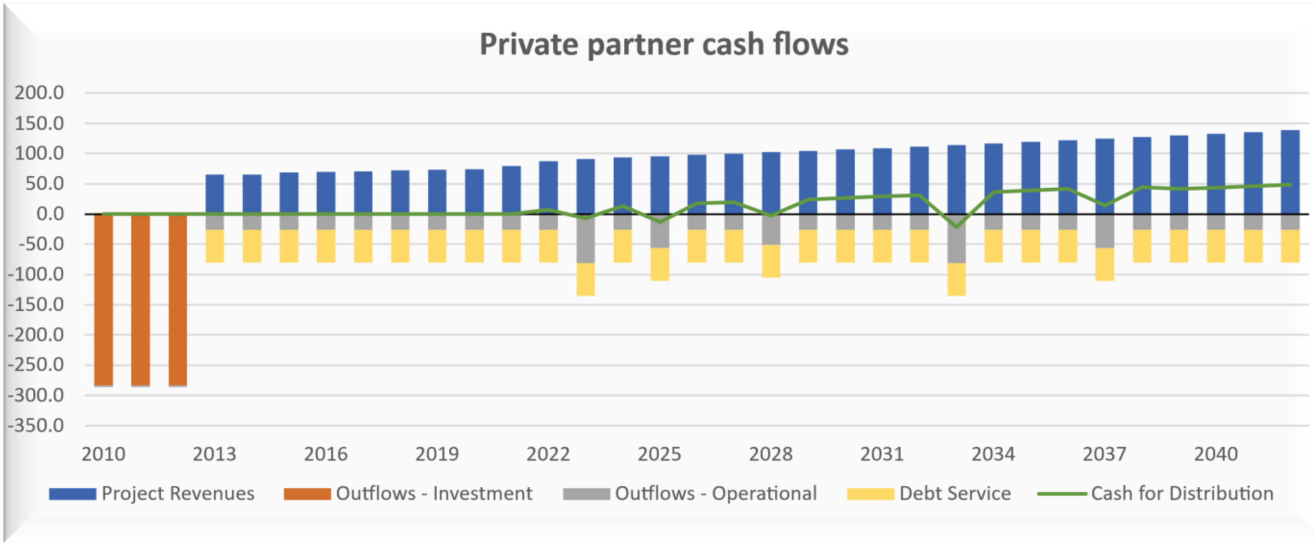

Figure 2: Eastern Motorway Project Cash Flows. *Inflation-linked availability payments steadily increase of the project's lifecycle. Heaviest cash outflows occur during the three-year construction phase of the project.*

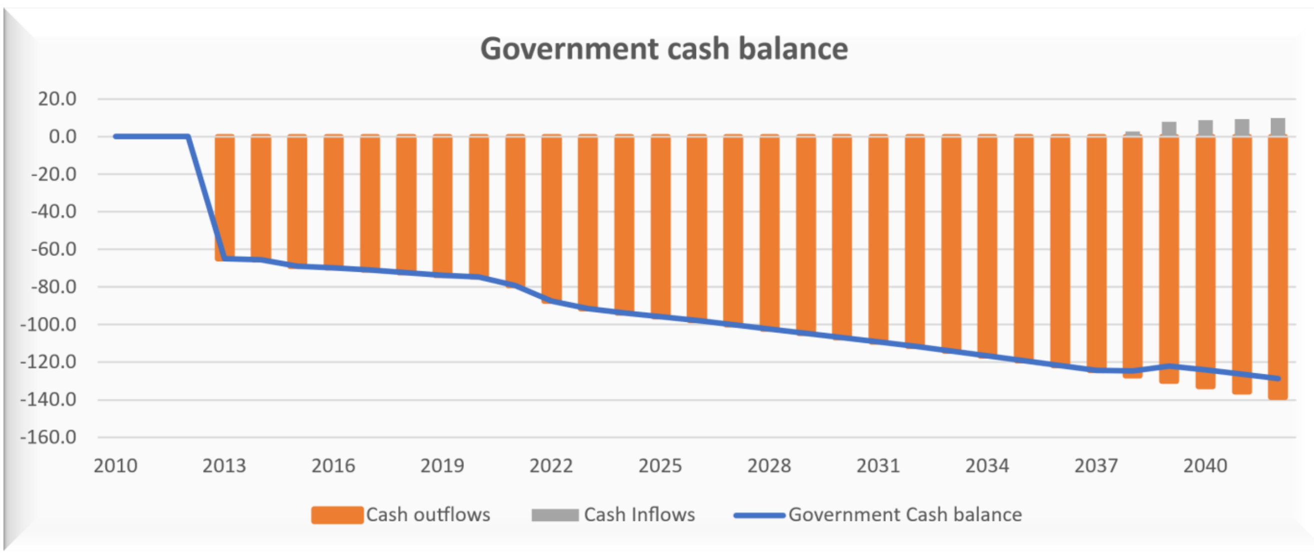

Figure 3: Eastern Motorway Impact on Government Cash Balance. *Annual outflows indicate availability payments. These are somewhat mitigated in the final years of the contract through tax revenues as the project becomes profitable.*

Most importantly, PFRAM 2.0 summarises the aggregate portfolio impact on fiscal balances, as illustrated in figures 4 and 5. The asset recognition profile generated by PFRAM 2.0 differs from the statistical treatment historically applied to Irish PPPs under ESA accounting rules. In practice, several Irish PPP assets remained off the general government balance sheet during the concession period and transferred formally to the State only at contract expiry. By contrast, PFRAM 2.0 adopts an economic-risk perspective, recognising government exposure progressively over the project lifecycle.

The model therefore reflects underlying fiscal exposure rather than formal national accounts classification.

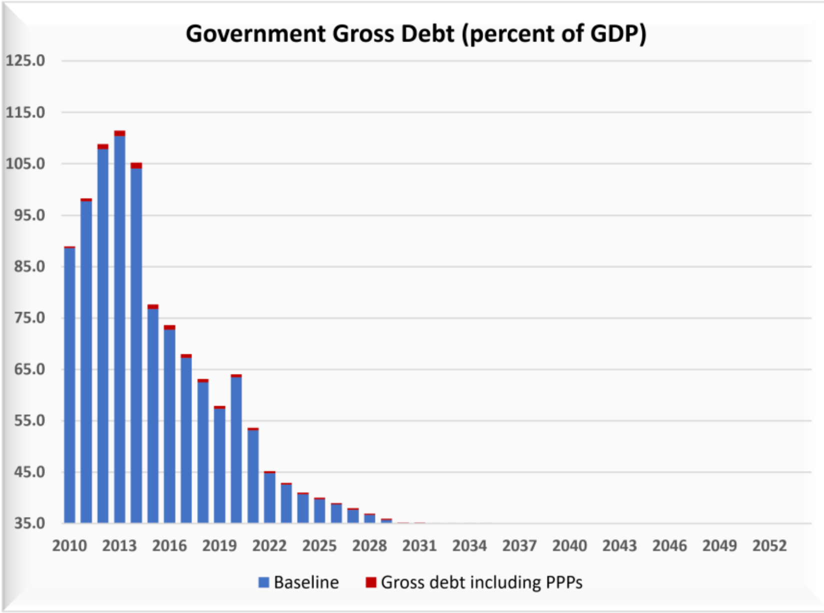

Figure 4: Government Gross Debt. *The PPP projects marginally impact debt in this constrained portfolio. Debt is largest as the project construction is completed between 2013 and 2014, decaying from there towards the long-term baseline assumption of 35%.*

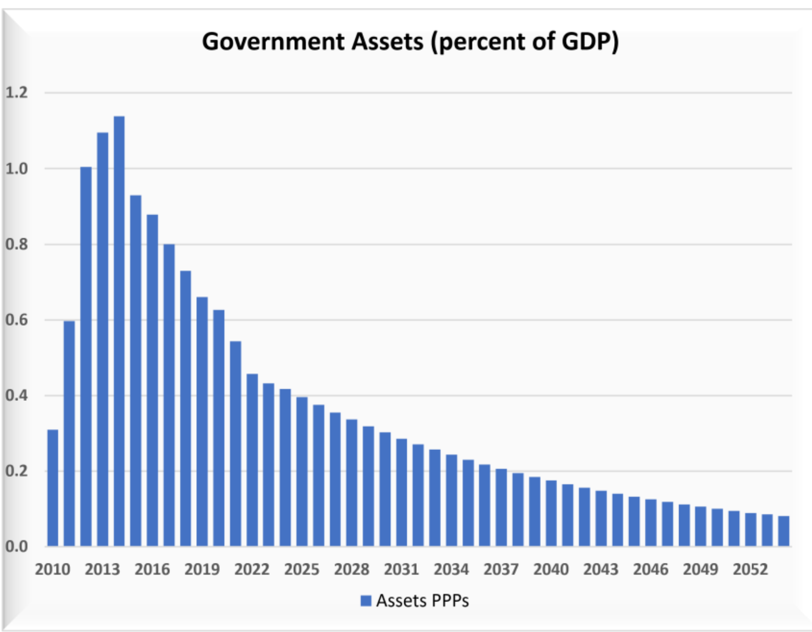

Figure 5: Government Assets. *Government assets are increased by the PPP portfolio under the IPSAS accrual accounting protocol. These assets decrease over time as the useful life of each infrastructure project is depleted.*

The baseline results show that the portfolio's central fiscal exposure is concentrated in the Eastern Motorway, reflecting its availability-payment structure. The Western Motorway and Southern Tunnel are primarily user-funded and therefore generate lower direct baseline fiscal costs. However, this should not be interpreted as lower risk overall. The Southern Tunnel contains contingent exposure through a demand guarantee, while the Western Motorway remains exposed to private-sector viability under traffic stress. The baseline therefore understates the fiscal relevance of these projects. This becomes clearer when a macro-economic shock is applied to the portfolio.

A 5% GDP shock between 2030 and 2035 is applied under PFRAM 2.0's assumption of independent risks. The model outputs the impact on government debt and contingent liabilities, as illustrated in figures 6 and 7.

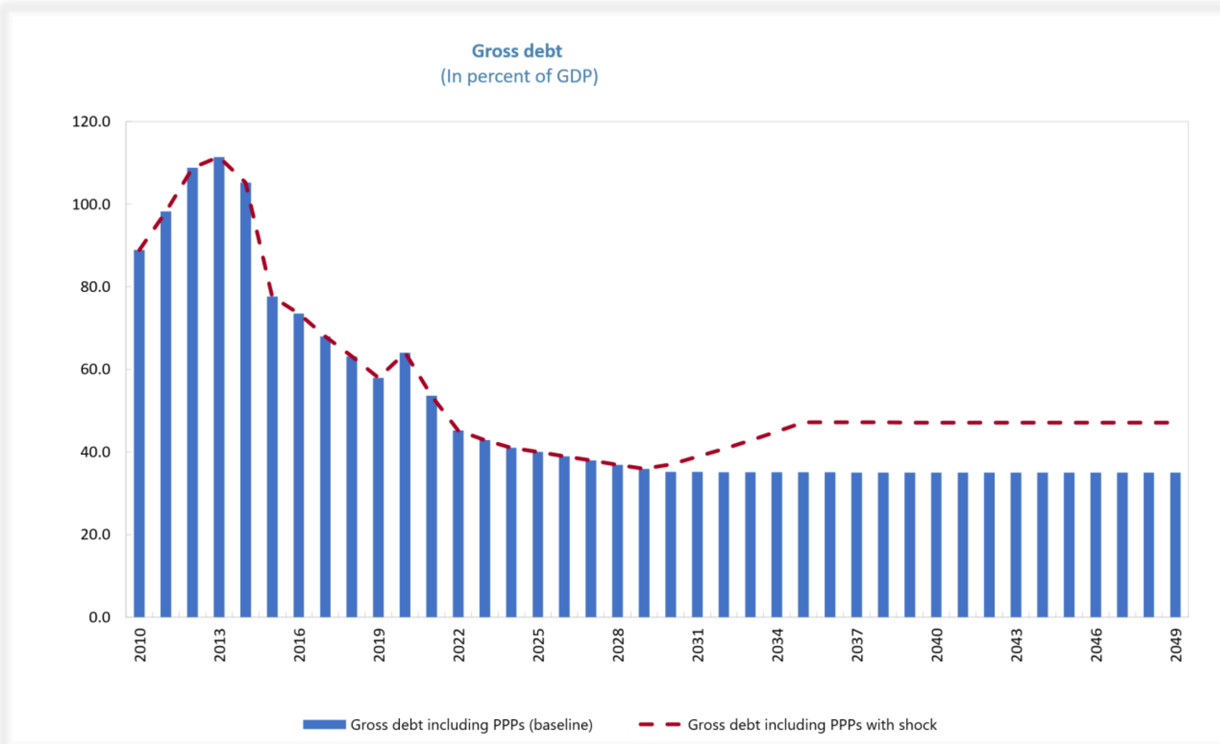

Figure 6: Gross Debt with Shock. *The debt ratio climbs as GDP falls and is sustained beyond the shock period as GDP remains lower than the base model assumptions.*

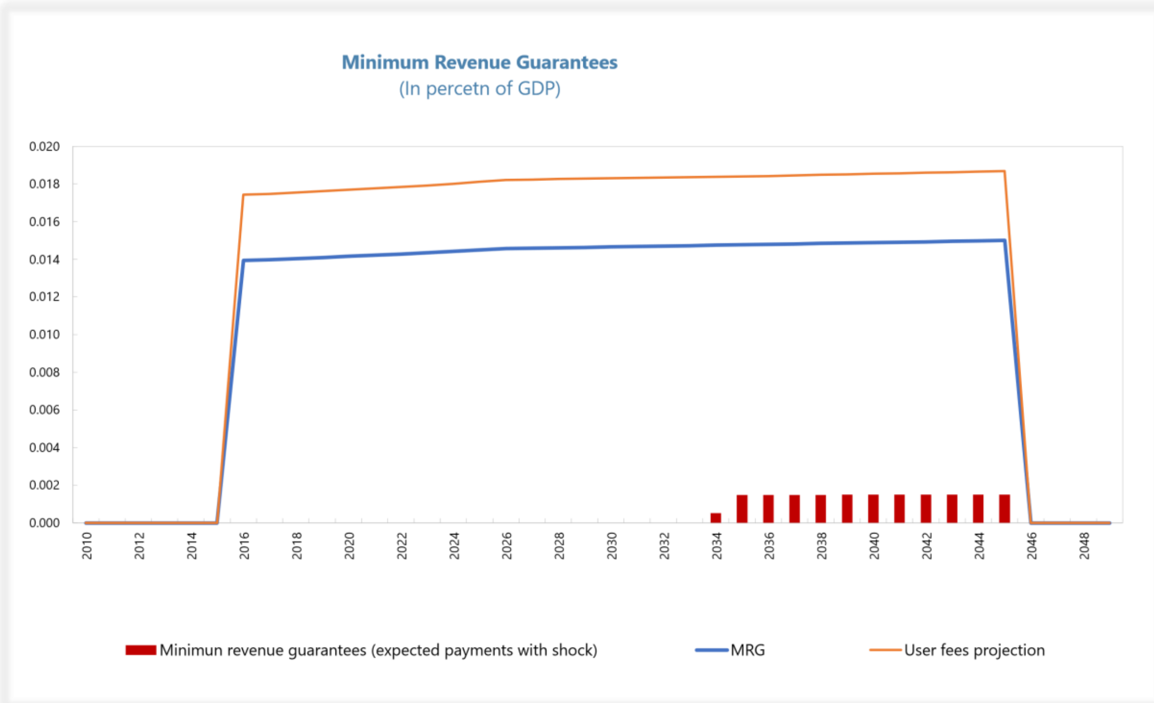

***Figure 7: Revenue Guarantees with Shock.** The fall in demand triggered the Southern Tunnel's contingent revenue support. Notably, payments continue after the shocked period. Demand begins to recover but remains below the base case and the project remains stressed.*

The shock triggered contingent liabilities in the Southern Tunnel as service demand dipped and government payments of approximately 0.0015% of GDP were observed. Although the payments are a small fraction of GDP in this case, it is straightforward to see how a larger portfolio, or a portfolio more exposed to demand risk would significantly impact government finances.

Utility of the straightforward GDP shock is however limited by the assumption of independent project risks.

Correlated Risk Results

The stochastic overlay examines the fiscal risk exposure of the three-project base portfolio in a more realistic stressed environment. The chosen stress scenario again begins with a 5% shock to demand over the 2030 to 2035 period. Developing beyond PFRAM 2.0, the demand shock is correlated with further shocks to project cost and financing to create a more realistic example.

These shocks translate into project losses based on the contractual structure of each project. The Western Motorway is not exposed to demand shocks directly and only generates a fiscal loss in extreme cases if the company becomes stressed and government step-in is required. As before, the Southern Tunnel generates losses when the contingent liability is triggered due to a fall in demand. The Eastern Motorway generates losses in response to cost and financing shocks, which affect availability payments under the assumed contractual model. Finally, to add a realistic layer of complexity, it is assumed that the Western Motorway and Southern Tunnel have a shared private partner. In this case, company stress generates fiscal losses simultaneously in both projects.

The shock is simulated 10,000 times to generate representative fiscal loss outcomes. Unlike the deterministic outputs of the PFRAM 2.0 model, the simulation permits analysis of the distribution of possible outcomes. Figure 8 provides a quantile summary of the simulated fiscal payments.

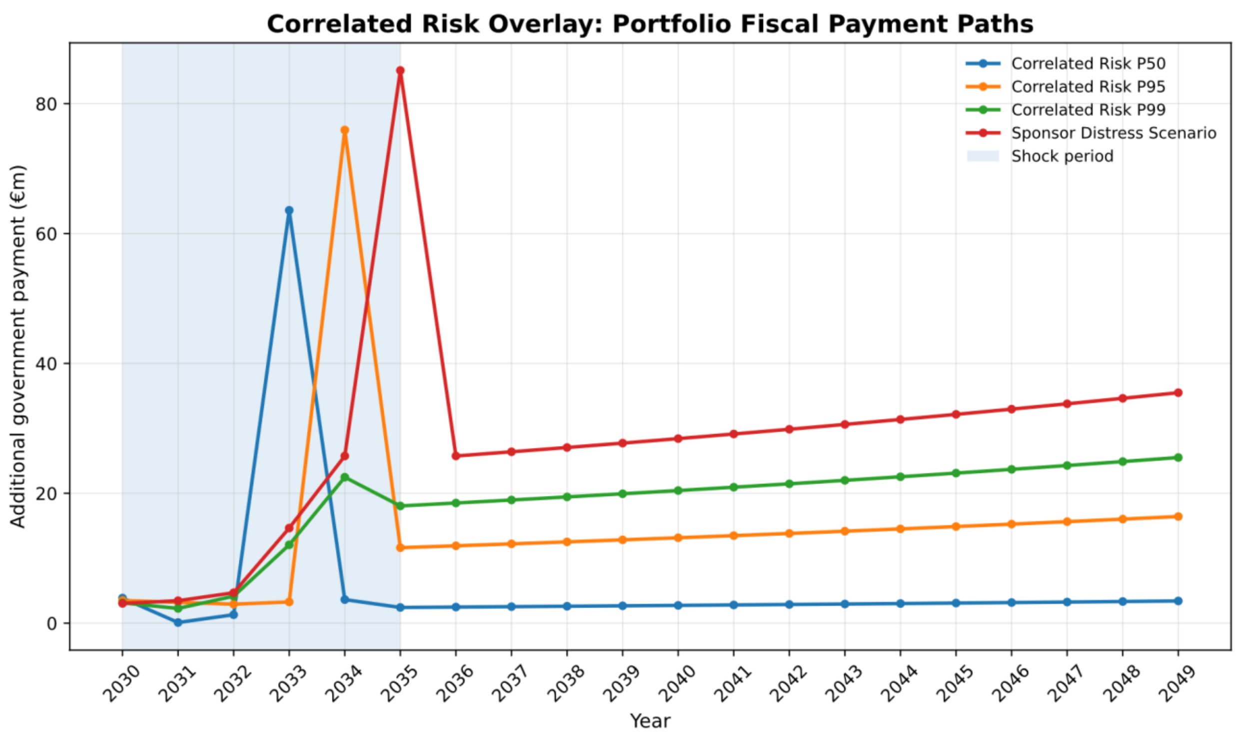

Figure 8: Summarised Correlated Simulation Results. ***Fiscal payment paths are illustrated at the 50th (median), 95th and 99th percentiles to examine the range of likely payment scenarios. Additionally, the Sponsor Distress Scenario extracts the highest simulated payment path. In each case, the spike in payments corresponds to government step in as one or more projects become distressed.***

These simulated payment paths can also be returned to the PFRAM 2.0 model to illustrate the shocks using the model's visualisation tools. Figures 9 and 10 illustrate the key fiscal summaries for the Sponsor Distress payment path.

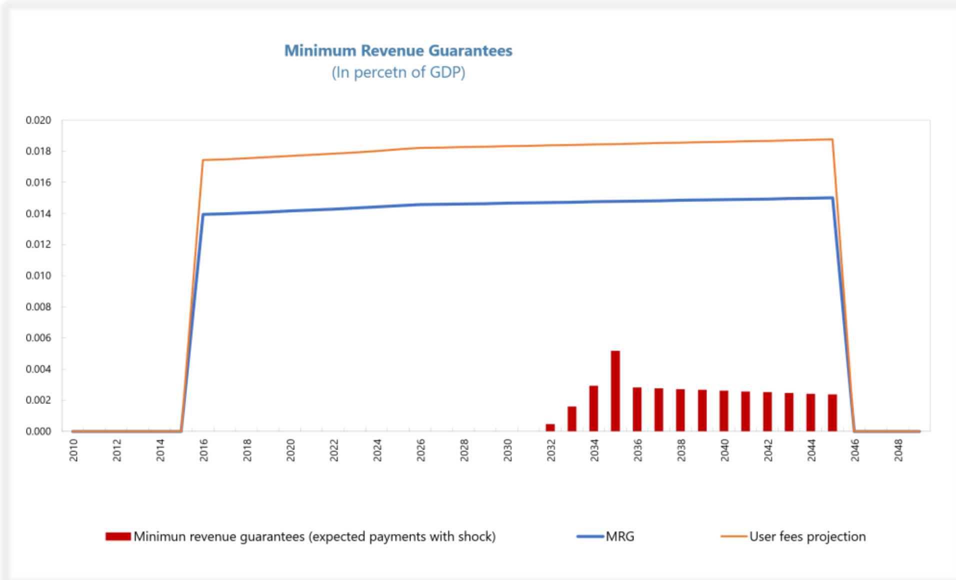

Figure 9: Government Payments incurred by the Sponsor Distress Scenario. *Payments are higher than in the base model, where the scenario was restricted to a 5% GDP shock.*

Figure 10: Debt Guarantee Payments incurred by the Sponsor Distress Scenario. *In addition to contingent revenue support, a one-off government payment was incurred in 2035 to support the private partner responsible for the Eastern Motorway and Southern Tunnel. As before, the payment is marginal with respect to GDP, given the synthetic portfolio's limited size.*

The Python simulation enabled a richer understanding of risk transmission than the base portfolio, and highlighted the vulnerabilities of projects not immediately exposed to demand risk. The script is highly flexible, and can be extended to incorporate copula and contagion analysis, which are beyond the scope of this proof of concept.

Diversification Analysis

In addition to modelling shared risks within the existing portfolio, the simulation framework permits analysis of new additions to the portfolio. The approach is light-weight, requiring only contract structure and capital expenditure details to build the risk framework. Table 3 summarised two potential additions to the portfolio, a fourth transport project and a social housing project. Both projects are of similar scale and government funded through availability payments. However, the addition of another transport project concentrates correlated risks within the portfolio, while the social housing project diversifies risks.

It is important to understand typical payment paths as well as worst case scenarios for the purposes of portfolio monitoring and diversification analysis. The same shock was applied to the portfolio as in the previous section, with fiscal outcomes summarised in figure 11.

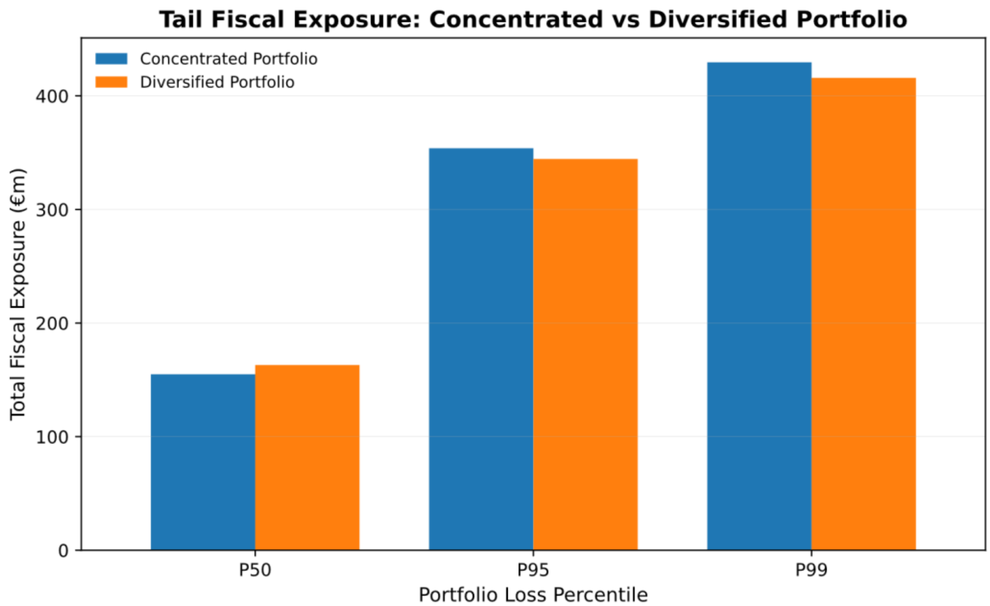

Figure 11: Shocked Fiscal Exposure of the Extended Portfolio. *The results illustrate the importance of understanding median (P50) scenario as well as worst case (P95 and P99) scenarios. The median scenario shows that the addition of another transport project (concentrated case) resulted in lower fiscal loss than the social housing (diversified case) addition. However, the worst-case scenarios both showed lower losses for the diversified portfolio.*

The diversification results indicate there is a balance between introducing new risks to the portfolio through new project types, and generating strong correlation of risks through a concentrated portfolio. Table 5 breaks these results down further. It is worth reiterating that the model results are sensitive to input assumptions, and the 3.1% tail reduction is directionally illustrative rather than a robust estimate.

Table 5: Portfolio Extension Analysis Summary. *Although median payments are higher, the diversified portfolio is less prone to extreme payments. This is reflected in a lower payment standard deviation, and a 99th percentile payment which is 3% lower than the concentrated case. Additionally, worst case scenario payments are significantly lower in the diversified portfolio.*

	Standard deviation (€m)	p50 payment (€m)	p99 payment (€m)	Max payment (€m)	Diversification tail reduction (P99)
Concentrated Transport Portfolio	110	155	430	625	
Diversified Portfolio	97	163	416	608	3.1%

Strategic Interpretation

The analysis illustrated the utility of the PFRAM 2.0 model for portfolio-level PPP monitoring. A simple form-based user interface makes modelling quicker than traditional financial analysis. The programme outputs useful non-technical visual summaries that can inform project selection, and on-going monitoring as revenues and costs materialise relative to the base case.

In addition to the government perspective, the model outputs equity IRRs for each project, allowing for basic assessment of private viability. This clarity is useful for the public partner during the contract negotiation phase of each project.

The core functionality of the PFRAM 2.0 model is sensitivity analysis. Monitoring the impact of GDP, Nominal Exchange Rate, and Inflation shocks informs reserve allocations at the budgetary level, as well as the long-term sustainability of the portfolio.

Risk modelling is limited to deterministic and independent macroeconomic shocks which are not typically realistic stress-testing assumptions. The stochastic layer constructed around the basic PFRAM 2.0 model addresses this and provides a more realistic assessment. The correlated risk analysis is useful to establish true worst-case scenarios in realistic stress-scenarios. Additionally, the output is simulation based, revealing the distribution of likely outcomes.

In a fully calibrated model, the stochastic framework could inform project selection. Consideration of diversification effects will allow the government to reduce volatility across the portfolio. Further to this, the framework can naturally be extended to investigate the benefits of sequencing projects to minimise risk correlations at each time step.

5. Policy Actions

The research suggests that Public-Private Partnerships are an important facilitator of Ireland's infrastructure strategy while the case study demonstrated that scaling PPPs in Ireland should not rely solely on project-level appraisal. The following pathway of actions translate the findings into practical policy reform.

Phase One: Analytic Capacity Building

Action 1: Create a central PPP portfolio risk function
It is recommended to establish a centralised risk function to manage aggregate PPP risk. The unit will have responsibility for maintaining a live PPP portfolio model.

Mechanism

The existing PPP oversight unit would be assigned and resourced as the owner of PPP portfolio risk. The Accelerating Infrastructure Report and Action Plan explicitly envisions the Department of Public Expenditure, Infrastructure, Public Service Reform and Digitalisation taking a central coordination role in infrastructure delivery, providing an existing mandate for this reform.

Portfolio modelling should begin with PFRAM 2.0 as a foundation and expanded by developing from the basic stochastic overlay used in this research. The complexity of risk analysis scales with the number of projects in the portfolio and developing a robust model is likely to take time and resources.

The main upfront cost of the proposed reforms is therefore resourcing the central oversight unit. A specialised risk modelling team will need to be developed, alongside strategy and coordination teams. Modelling actions are low cost. PFRAM 2.0 is a publicly available analytical tool and can be used by government departments without licensing fees or proprietary software costs. A stochastic risk layer can be implemented with open-source coding software, such as Python.

Limitations and Trade-offs

Buy-in to the Accelerating Infrastructure Report and Action Plan is strong, reflecting a consensus that Ireland's approach to infrastructure delivery must be reformed. Establishing a centralised PPP risk unit should be viewed as part of this strategy. The risk here is pushback over 'government-by-agency', therefore it is crucial that the new risk function is embedded within the exiting PPP unit and chaired by the minister for Public Expenditure, Infrastructure, Public-Service Reform and Digitalisation to ensure a direct line of communication.

Phase one is largely technical and faces practical constraints. Reliable modelling requires accurate data that may not be centrally available. Data sharing arrangements will need to be developed with the sponsoring agencies to mitigate this risk. There is also model risk and uncertainty, particularly around the stochastic elements. A year-long timeframe is recommended to iteratively improve and build confidence in modelling outputs, however this may need to be extended if technical issues are encountered.

Phase Two: Go-Live

Action 2: Embed portfolio stress-testing within the Public Spending Code

Reform the Public Spending Code to include Portfolio stress-testing as part of the PPP procurement process.

Action 3: Improve fiscal disclosure

The modelling framework creates an opportunity to increase fiscal disclosures related to the portfolio. Private-sector involvement in large infrastructure projects can generate public backlash, and transparent reporting will help mitigate this risk.

Mechanism

Phase two makes the analysis consequential by integrating it within the formal appraisal process. Within the Public Spending Code, portfolio stress-testing should be sequenced between construction of a Public Sector Benchmark and project tendering[2], as illustrated in figure 12. The marginal effect of new PPPs on total fiscal exposure should be modelled, expanding beyond standalone Value for Money assessment. Annual accounts and budget documents should incorporate a PPP portfolio risk section, drawing on PFRAM 2.0's non-technical visual summaries.

[2] The latest version of Ireland's Public Spending Code is available here.

Figure 12: Proposed Public Spending Code Adjustment. *Portfolio Analysis follows the existing Value for Money and Public Sector Benchmarking exercises. The portfolio analysis informs an Impact Assessment, which is incorporated into the third phase of reform.*

Limitations and Trade-offs

Embedding portfolio analysis within the Public Spending Code adds an additional layer of complexity that may slow the appraisal process. For this reason, the second phase of reform initially introduces portfolio modelling without influencing project selection. This will give the modelling time to establish assessment processes without delaying live projects.

Increasing the transparency of fiscal reporting is a governance improvement but also entails risk. A step-change in PPP reporting detail may increase both public and political scrutiny of the existing PPP portfolio. As such, beginning with a subset of visual portfolio summaries is recommended to manage this risk.

Phase Three: Strategic Integration

Action 4: Expand remit of the oversight unit to include project selection

The central oversight unit must have the authority to influence project selection based on portfolio analysis. This recommendation is crucial to align projects with the government's wider strategic vision.

Action 5: Use diversification and sequencing rules in project selection

The reformed Public Spending Code should consider diversification effects and project sequencing as part of the stress-testing process. This will protect the long-term sustainability of fiscal obligations and create the conditions necessary to scale the portfolio.

Mechanism

With the portfolio unit in place and analysis embedded within the appraisal framework, phase three focuses on using that analysis to guide the pipeline of projects.

A comply-or-explain framework is recommended to balance oversight with departmental autonomy. For each proposed project, portfolio analysis would be conducted to assess the marginal impact on fiscal exposure, incorporating diversification and sequencing considerations. Guided by this analysis, the centralised risk unit would produce an impact assessment outlining the key portfolio risks associate with the project. The impact assessment would be formally responded to by the sponsoring department minister, either adjusting the project timeline and design to mitigate risk or explaining why the project's benefits justify taking on the added risk. The minister retains final decision-making here, ensuring autonomy and transparency.

Limitations and Trade-offs

This phase entails the highest level of political risk, because it affects departmental autonomy. The trade-off here is between greater fiscal prudence and executive flexibility. The comply-or-explain framework retains a level of departmental autonomy to balance this risk. The central PPP unit should exercise influence in a restrained manner, presenting advice rather than orders. Additionally, the expertise developed within the central unit can be deployed to provide consultation to sponsoring bodies.

There is also a broader political economy risk specific to Ireland's coalition government structure. Where infrastructure projects carry electoral significance, ministers may face constituent pressure that conflicts with portfolio-level fiscal logic. The comply-or-explain mechanism is designed to mitigate this risk. By requiring ministers to publicly justify deviations from portfolio advice, it creates accountability without removing democratic discretion.

This phase also entails procedural risk, as the addition of an impact assessment slows the appraisal process. This is particularly challenging given the current governmental push for efficiency. A conservative approach might be justified here, requiring impact assessments only where a project is above a specified capital threshold. Embedding the risk function within the existing central PPP unit is another mitigant here that avoids building new institutional bureaucracies.

Workflow Diagram

Figure 13 provides an indicative timeline to implement the phased pathway. Technical complexity, procedural reform, and departmental buy-in are the key risks necessitating a phased approach.

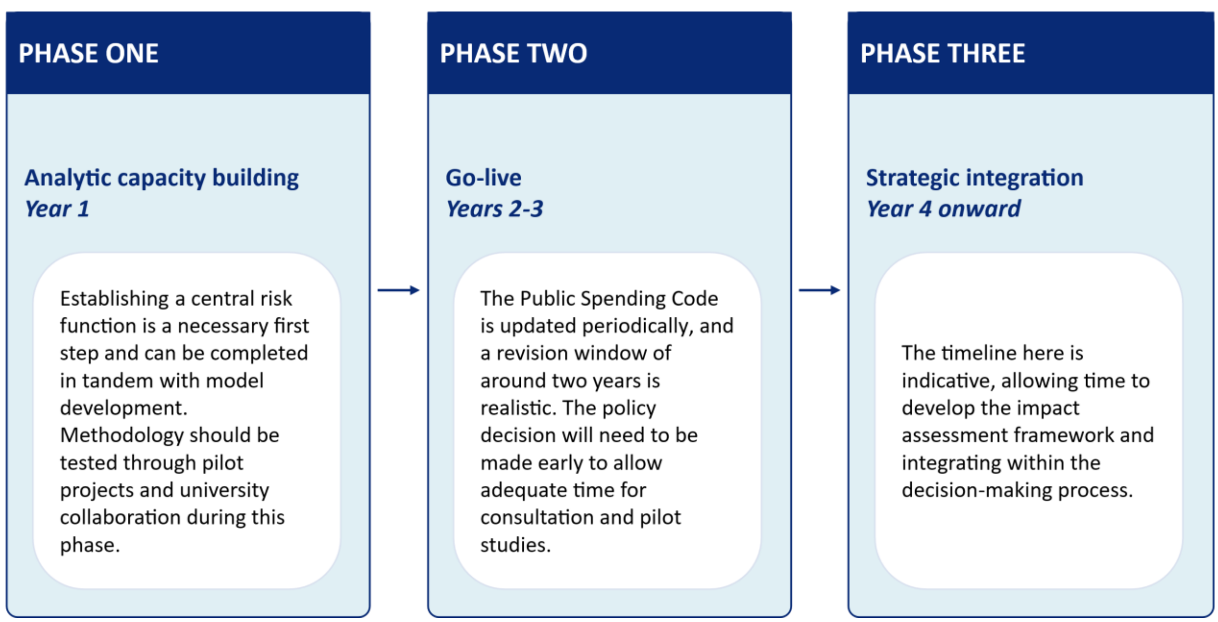

Figure 13: Indicative Reform Workflow.

Conclusions

The scale of Ireland’s public infrastructure pipeline demands policy reform. The existing approach to infrastructure delivery is fractured, with projects frequently delayed and over-budget. To address these challenges, the government is focused on building public sector expertise and enhancing coordination across projects.

While this report finds that coordination brings significant benefits, it argues that reduced private sector involvement may undermine efficient delivery. Public-Private Partnerships are an established framework to incorporate private sector efficiency into public infrastructure delivery, while reducing reliance on expensive ad hoc consultancy. PPPs will be important to delivering investment at the scale the government envisages.

The central finding of the report is that scaling PPP use requires centralised portfolio management. The existing framework largely assesses project risk on an individual basis, which underestimates aggregate fiscal exposure when shared risks are accounted for. Additionally, assessing the contribution of new projects to the existing portfolio enables policymakers to align projects with wider strategic goals, optimise their sequencing, and assess their contribution to portfolio risk diversification.

Analysis of a synthetic PPP portfolio stylised on Ireland's existing PPPs illustrated the potential benefits of portfolio monitoring and the structural logic underpinning the proposed approach. The PFRAM 2.0 model is a ready-made tool which can be adopted by policymakers in Ireland and can be extended to incorporate realistic stress-testing and diversification analysis.

The report findings translate into a clear policy recommendation: Ireland needs a central PPP unit with the capacity to model portfolio risk and select projects strategically. The mandate of the existing unit must go beyond supervisory support to active participation and oversight.

Modelling in this report has been simplified to maintain analytical clarity. Additionally, limited publicly disclosed details of Ireland’s PPPs prevented full replication of the projects within the PFRAM 2.0 model. The model itself is limited in its ability to reflect complex project debt structuring and conduct realistic sensitivity analysis. The central lesson is that PPPs necessarily entail risk, and we must endeavour to manage it effectively.

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Appendix – Modelling Assumptions

Table 6: Macro Assumptions in the PFRAM Model. *The macroeconomic baseline is calibrated to Ireland's observed macro-fiscal position over 2010–2024, gathered from the Central Statistics Office National Account Data. From 2025 onward, the spreadsheet applies simple deterministic projection.*

Parameter	2024 Value	Projection rule from 2025 onward
Nominal GDP	€538 billion	4.50% annual growth
Real GDP	€396 billion	2.25% annual growth
Inflation	135.8 (index value relative to 2014)	2.20% annual growth
Government revenue	€128 billion	4.5% annual growth
Primary balance	€30 billion	4.5% annual growth
Cash balance	€24.5 billion	4.5% annual growth
Government gross debt	€219 billion	Stabilises at 35.0% of GDP from 2030 onward

Table 7: Project Assumptions for the Baseline Portfolio.

	Western Motorway	Southern Tunnel	Eastern Ring Road
Contract type	DBFOM	DBFOM	DBFOM
Funding	User-funded	User-funded with demand guarantee	Government-funded
Concession period	2010–2040	2012–2046	2010–2043
Capex	€700m	€950m	€850m
Construction Length	3 years	4 years	3 years
Debt Gearing	85%	88%	90%
Blended Interest Rate	5.80%	6.10%	5.20%
Revenue Model	Initial toll revenue €29.2m growing in line with GDP.	Initial toll revenue of €49.7m growing in line with GDP. Guaranteed up to 80% of base forecast.	Availability payment of €65m per year indexed to inflation.
Demand profile	Traffic demand elastic with GDP	Traffic demand elastic with GDP	Not demand-driven
Operational costs (O&M)	€2m/year during construction, €8m during operation. Maintenance cost €6m/year.	€2m/year during construction, €18m during operation. Maintenance cost €7m/year.	€2.5m/year during construction, €16m during operation. Maintenance cost €10m/year.

Table 8: Correlated Risk Simulation Setup. *The correlated-risk layer simulates 10,000 fiscal-loss paths for the base three-project portfolio over 2030–2049. During the 2030–2034 shock period, demand, cost and financing shocks are drawn jointly using a correlation structure below. Shared contractor distress is modelled as a one-off event linked to severe transport-revenue stress and applies to the Western Motorway and Southern Tunnel only.*

Assumption	Model specification
Simulation period	2030–2049
Shock period	2030–2034
Number of simulations	10,000
Simulated risk factors	Demand shock; cost shock; financing shock
Base annual demand shock	5.0%, with stochastic variation
Base annual cost shock	3.0%, with stochastic variation
Base annual financing shock	1.5%, with stochastic variation
Demand–cost correlation	0
Demand–financing correlation	0.35
Cost–financing correlation	0.45
Shared contractor distress trigger	Cumulative stressed transport revenue falls below 80% of baseline revenue
Shared contractor distress probability	25% during 2030–2034 if severe revenue stress is present; 10% during 2035–2037 if stress persists
Shared contractor distress frequency	One-off event only
Projects exposed to shared contractor distress	Western Motorway and Southern Tunnel

Table 9: Correlated Risk Transfer Mechanisms by Project. ***This table shows how the simulated shocks are translated into additional government payments for each project in the base PPP portfolio. The Western Motorway is a pure concession and only generates a fiscal payment if the shared contractor distress event occurs. The Southern Tunnel is exposed to both revenue-guarantee payments and the same shared contractor distress event. The Eastern Ring Road is an availability-payment PPP and transmits cost and financing shocks into higher public payments, but it is not assigned to the shared contractor distress channel.***

	Western Motorway	Southern Tunnel	Eastern Ring Road
Direct demand-risk transfer	No	Yes	No
Demand-risk mechanism	Not applied	Shortfall between 80% of baseline revenue and stressed revenue	Not applied
Cost-risk transfer	No	No	Yes
Cost-risk mechanism	Not applied	Not applied	cost shock x baseline availability payment
Financing-risk transfer	No	No	Yes
Financing-risk mechanism	Not applied	Not applied	0.40 x financing shock x baseline availability payment
Shared contractor / counterparty exposure	Yes	Yes	No
Shared contractor / counterparty fiscal effect	One-off €40m payment if distress event occurs	One-off €40m payment if distress event occurs	Not applied

Table 10: Diversification Assumptions. *The diversification analysis compares two alternative incremental availability-payment PPPs of equal baseline scale. Both portfolio variations still include the original three-project base portfolio and its existing shared counterparty-risk structure.*

	Concentrated transport PPP	National social housing PPP
Funding	Availability PPP	Availability PPP
Baseline availability payment	€70m in 2030	€70m in 2030
Payment indexation	2.5% p.a.	2.5% p.a.
Demand-risk transmission	Additional fiscal loss equals 0.35 × annual demand shock × baseline payment	Not modelled
Cost-risk transmission	Additional fiscal loss equals 1.00 × cost shock × baseline payment	Additional fiscal loss equals 0.85 × cost shock × baseline payment
Financing-risk transmission	Additional fiscal loss equals 0.40 × financing shock × baseline payment	Additional fiscal loss equals 0.35 × financing shock × baseline payment